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EEE 322 — Past Questions & Materials

by Renance

Course: EEE 322

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 4

Levels: 300

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Paper 1 of 4 — EEE 322 - 2024/2025 - Second Semester - Electrical Power Systems Principles

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EXAMINATION: Second Semester

COURSE CODE: EEE 322

SESSION: 2024/2025

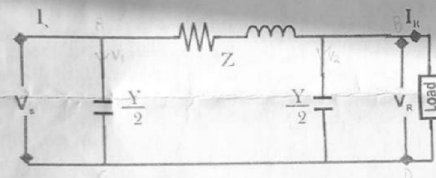
TITLE: Electrical Power Systems Principles

DEGREE: B.ENG



TIME ALLOWED: 3 Hours

INSTRUCTION: Answer any FIVE questions

QUESTION 1Figure 1: NOMINAL- π MEDIUM TRANSMISSION LINE

- i). Using nominal- π medium transmission line shown in Figure 1, show that for the generalized circuit constants, **10marks**

$$A = D = 1 + \frac{YZ}{2}, \quad B = Z, \text{ and } C = Y + \frac{Y^2 Z}{4}$$

- ii). What are the units of the generalized constants of a transmission line **4marks**

QUESTION 2

- a). Find the following for a single circuit transmission line delivering a load of 50MVA at 110kV, and pf. 0.8 lagging: **10marks**

- sending end voltage,
- Sending end current
- sending end power

$$S = \text{kVA} \quad P = \text{kW} \quad \text{pf} = \cos \phi$$

$$A = 1 + \frac{YZ}{2} \quad B = Z \quad C = Y + \frac{Y^2 Z}{4}$$

$$V_s = V_r + IR$$

$$I_s = I_r + I_C$$

$$P_s = P_r + P_{\text{loss}}$$

2

iv. The transmission efficiency
 Given $A = D = 0.98$ < 3; $B = 110$ < 75 degrees ohm; $C = 0.0005$ < 80 degree mho
 b). What are the causes of voltage drop and line loss in a transmission line. **4marks**

QUESTION 3

3a. An overhead transmission line conductor having parabolic configuration weighs 1.925kg per meter length, area of cross section 2.2cm^2 and an ultimate strength of 8000 kg/cm^2 , when erected between supports 600m apart and having 15m difference in height, determine the vertical sag from the taller of the two supports which must be allowed so that the factor of safety shall be 5 with the wire loaded due to 1kg of ice per meter and no wind pressure. **9marks**

3b. List at least five factors that affect sag in a transmission line **5marks**

QUESTION 4

4a. A 400V, 50Hz, 3-phase line delivers 200kw at 0.8 pf lagging. It is desired to raise the line power factor to unity by installing shunt capacitors. Calculate the capacitance of each unit if they are connected in (i) star and (ii) delta **7marks**

4b Define the following: **3marks**

I) demand factor, II) load factor and III) diversity factor

4c. What purposes are served by tariff? **5marks**

QUESTION 5

5a. The Ugheli generating station has a maximum demand of 20MW, a load factor of 60 percent, plant capacity factor of 48 percent and a plant use factor of 80 percent. Find:

I) The daily energy consumed **3marks**

II) The reserve capacity of the plant, **3marks**

III) The maximum demand that could be produced daily, if the plants were running all the time and, **2marks**

IV) The maximum energy that could be produced daily, if the plants were running fully loaded. **2marks**

5b. What are the significance of load factor and the diversity factor. **4marks**

QUESTION 6

6a. Develop and explain a model that can represent electricity generation from various sources. **8marks**

6b. What are the limitations of electricity generation by

Paper 2 of 4 — EEE 322 - 2024/2025 - Second Semester - Electrical Power Systems Principles

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EXAMINATION: Second Semester

COURSE CODE: EEE 322

SESSION: 2024/2025

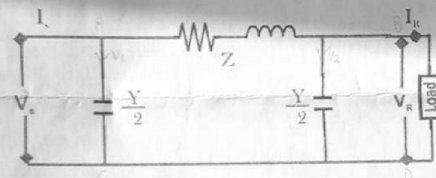
TITLE: Electrical Power Systems Principles

DEGREE: B.ENG



TIME ALLOWED: 3 Hours

INSTRUCTION: Answer any FIVE questions

QUESTION 1Figure 1: NOMINAL- π MEDIUM TRANSMISSION LINE

- i). Using nominal- π medium transmission line shown in Figure 1, show that for the generalized circuit constants, **10marks**

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- ii). What are the units of the generalized constants of a transmission line **4marks**

QUESTION 2

- a). Find the following for a single circuit transmission line delivering a load of 50MVA at 110kV, and pf. 0.8 lagging: **10marks**

- sending end voltage,
- Sending end current
- sending end power

$$S = \text{kVA} \quad P_f = 0.8$$

$$P = \text{kW}$$

$$Q = \text{kVAR}$$

$$A = 1 + \frac{YZ}{2} \quad B = Z \quad C = Y + \frac{Y^2 Z}{4}$$

$$V_s = V_r + IR$$

$$I_s = I_r + I_C$$

$$P_s = P_r + P_C$$

2

iv. The transmission efficiency
 Given $A = D = 0.98$ < 3; $B = 110$ < 75 degrees ohm; $C = 0.0005$ < 80 degree mho
 b). What are the causes of voltage drop and line loss in a transmission line. **4marks**

QUESTION 3

3a. An overhead transmission line conductor having parabolic configuration weighs 1.925kg per meter length, area of cross section 2.2cm^2 and an ultimate strength of 8000 kg/cm^2 , when erected between supports 600m apart and having 15m difference in height, determine the vertical sag from the taller of the two supports which must be allowed so that the factor of safety shall be 5 with the wire loaded due to 1kg of ice per meter and no wind pressure. **9marks**

3b. List at least five factors that affect sag in a transmission line **5marks**

QUESTION 4

4a. A 400V, 50Hz, 3-phase line delivers 200kw at 0.8 pf lagging. It is desired to raise the line power factor to unity by installing shunt capacitors. **7marks**
 Calculate the capacitance of each unit if they are connected in (i) star and (ii) delta

4b Define the following: **3marks**
 I) demand factor, II) load factor and III) diversity factor

4c. What purposes are served by tariff? **5marks**

QUESTION 5

5a. The Ugheli generating station has a maximum demand of 20MW, a load factor of 60 percent, plant capacity factor of 48 percent and a plant use factor of 80 percent. Find:
 I) The daily energy consumed **3marks**
 II) The reserve capacity of the plant, **3marks**
 III) The maximum demand that could be produced daily, if the plants were running all the time and, **2marks**
 IV) The maximum energy that could be produced daily, if the plants were running fully loaded. **2marks**

5b. What are the significance of load factor and the diversity factor. **4marks**

QUESTION 6

6a. Develop and explain a model that can represent electricity generation from various sources. **8marks**

6b. What are the limitations of electricity generation by

Paper 3 of 4 — EEE 322 ELECTRIC POWER SYSTEM PRINCIPLES

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EXAMINATION: Second Semester

COURSE CODE: EEE 322

SESSION: 2023/2024

TITLE: Electric Power System Principles

DEGREE: B.ENG

TIME ALLOWED: 3 Hours



INSTRUCTION: Answer any FIVE questions

1. A short 3-phase transmission line of parameters $R = 0.4\Omega$ is delivering 2000kVA to a load at a p.f of 0.8 lagging at the receiving end of the line. If the load voltage is 3000V,
 - I. Determine the performance of the transmission line 10Marks
 - II. What is the significance of your result in (I) 4Marks
2. An overhead transmission line at a river crossing is supported from two towers at heights of 25m and 75m above the water level. The horizontal distance between the towers is 250m. If the required clearance between the conductor and the water mid-way between the towers is 45meters and if both the towers are on the same side of the point of maximum sag of the parabolic configuration. Take the weight of the conductor as 0.7kg/m.
 - I. Show the diagrammatic representation
 - II. Obtain the stringing tension in the conductor.
- 3a. What are the purpose served by tariff charged by the utility company in charge of your district? 4marks
- 3b. A consumer takes a steady load of 300kw at a lagging power factor of 0.7 for 3,000 hours a year. The tariff is 1300 naira per kVA of maximum demand per annum. The annual cost of the phase advancing plant is 130 per kVAR. How much is the annual saving if the pf of the load is improved? 10marks
4. The load on Delta IV power plant on a typical day is shown in Table 1.

Time (hours)	12-5 AM	5-9 AM	9-6 AM	6-10 AM	10-12 AM
Load (MW)	20	40	80	100	20

 - I. Develop chronological load model
 - II. Develop load duration model
 - III. Obtain the load factor of the plant and the energy supplied by the plant in 2 hours
5. A 666.66kw, 50hz, 400V, 3-phase load has a power factor of 0.8 lagging. It is desired to improve the power factor to 0.95 lagging by using a delta-connected bank of 2200V capacitor. Determine,
 - I. Capacitance of each capacitor
 - II. kVA rating of capacitor bank and transformer.

6a. Mention the six components of a power system

6marks

6b. What are the sources of electrical energy generation

3marks

6c. What are the challenges faced by the installation of solar photovoltaic power plant in some state in Nigeria.

5marks

7. A 50-Hz, 3-phase transmission line is 200km long. It has a total series impedance of $(35 + j140)\Omega$ and shunt admittance of $930 \times 10^{-6} \angle -90^\circ \text{ S}$. It delivers 40MW power at 220kV with 0.9pf lagging. Find the magnitude of the sending end voltage using nominal pi method of the line

10marks

⑤ The efficiency is high due to the lagging of the power factor at the receiving end

5) Capacitance of each capacitor

$$I_c = \frac{\text{Power}}{\sqrt{3} \times \text{Voltage} \times \cos \phi}$$

$$= \frac{600.66 \times 10^3}{\sqrt{3} \times 400 \times 0.8} = 1202.8 \text{ A}$$

for initial current I_1

$$I_1 = I_c (\cos \phi_1 - j \sin \phi_1)$$

$$I_1 = 1202.8 (0.8 - j0.6)$$

$$= 962.24 - j721.68$$

⑦



Paper 4 of 4 — EEE 322 Power Systems Principles

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATION: Second Semester

SESSION: 2017/2018

DATE: September, 2018.



COURSE CODE: EEE 322

COURSE TITLE: Power system principles

TIME ALLOWED: 3 Hours

INSTRUCTION: Attempt five (5) questions only.

Question 1

- (a) List five (5) sources of energy and list the factors responsible for the comparison of these sources of energy? **(3 Marks)**

(b) State five (5) advantages and disadvantages of hydroelectric power station. A hydro-electric generating station is supplied from a reservoir of capacity 5×10^6 cubic meters at a head of 200 meters. Find the total energy available in kWh if the overall efficiency is 75%. **(7 Marks)**

- (c) List the schematic arrangement of a nuclear power plant. A diesel power station has the following data:

Fuel consumption/day = 1000 kg

Units generated/day = 4000 kWh

Calorific value of fuel = 10,000 kcal/kg

Alternator efficiency = 96%

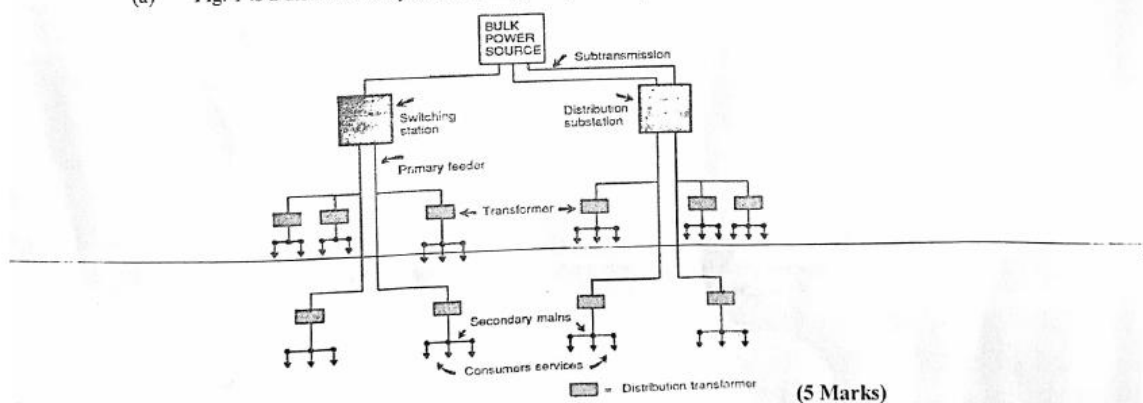
Engine mech. efficiency = 95%

Estimate (i) specific fuel consumption, (ii) overall efficiency, and (iii) thermal efficiency of engine **(6 Marks)**

- (d) Briefly explain the barrages and dynamic tidal power. **(4 Marks)**

Question 2

- (a) Fig. 1 is a distribution system showing components parts, briefly explain the system. **(5 Marks)**

**Fig. 1**

- (b) A power station has the following load:
 Residential lighting load: maximum load = 1,000kW

Load factor = 20%

Diversity between customers = 1.3

Commercial load: maximum load = 2,000kW

Load factor = 30%

Diversity between consumers = 1.1

Industrial load: maximum demand = 5,000kW
 Load factor = 80%
 Diversity between consumers = 1.2
 Overall diversity factor may be taken as 1.4

Find (i) maximum demand on the system

(ii) Daily energy consumption of each type of load and total energy consumption.

(iii) Connected load of each type, assuming the demand factor of each is 100%. (7.5 Marks)

(c) Briefly explain these types of loads (i) Domestic load (ii) Commercial load (iii) Industrial load (iv) Agricultural loads. (4 Marks)

(d) It has been desired to install a diesel power station to supply power in a suburban area having the following particulars:

(i) 1000 houses with average connected load of 1.5 kW in each house. The demand factor and diversity factor being 0.4 and 2.5 respectively.

(ii) 10 factories having overall maximum demand of 90 kW.

(iii) 7 tube wells of 7 kW each and operating together in the morning.

The diversity factor among above three types of consumers is 1.2. What should be the minimum capacity of power station? (3.5 Marks)

Question 3

- (a) List the advantages of interconnecting a grid system? (3 Marks)
 (b) State the classification of a distribution system? Fig 2. Is a secondary distribution system, briefly discuss it?

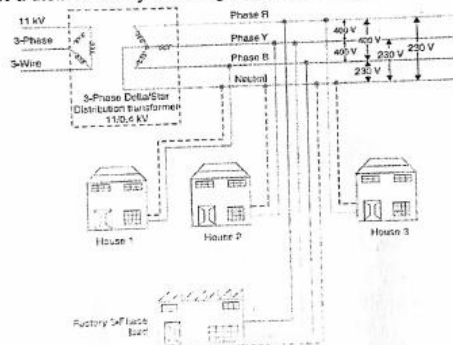


Fig. 2

(9 Marks)

- (c) Fig. 3 is a type of connection scheme in a distribution system, discuss briefly the scheme, stating advantages.

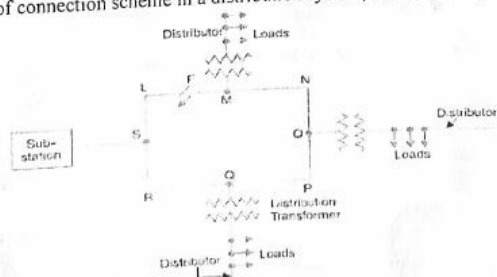


Fig.3

(5 Marks)

- (d) List the types of phase connections in an alternating current (A.C) distribution? (3 Marks)

Question 4

- (a) What is a short transmission line and state the parameters governing the performance of the transmission lines. (4 Marks)

- (b) In the determination of generalized constants for transmission line, derive the equation for medium lines using the Nominal T method. (7 Marks)
- (c) Define the term ferranti effect and stating the factors affecting corona effects (3 Marks)
- (d) A 3 – phase line, 3km long delivers 3,000kW at a power factor of 0.8 (lagging) to a load. If the voltage at supply end is 11kV, determine the voltage at the load end and the efficiency of transmission. The resistance and reactance per km of each conductor are 0.4Ω and 0.8 respectively. (6 Marks)

Question 5

- (a) Briefly discuss the single line diagram, impedance diagram and the reactance diagram representation of power system. (3 Marks)
- (b) State the advantages and disadvantages of per unit (P.U) system. A single phase 9.6kVA, 500V/1.5kV transformer has an impedance of 1.302Ω with respect to primary side. Find its per unit impedance with respect to primary and secondary sides. (6 Marks)
- (c) A 300MVA, 20kV three-phase generator has a subtransient reactance of 20%. The generator supplies a number of synchronous motors over 64km transmission line having transformers at both ends shown in Fig 4. The motors, all rated 13.2kV, are represented by just two equivalent motors. Rated inputs to the motors are 200MVA and 100MVA for M_1 and M_2 respectively. For both motors $X'' = 20\%$. The three phase transformer T_1 is rated 350MVA, 230/20kV with leakage reactance of 10%. Transformer T_2 is composed of three single phase transformers each rated 127/13.2 kV, 100MVA with leakage reactance of 10%. Series reactance of the transmission line is $0.5\Omega/\text{km}$. Draw the impedance diagram, with all impedances marked in per unit. Select the generator rating as base in the generator circuit. (8 Marks)



Fig. 4

- (d) List the advantages of using shunt capacitors in a transmission line. (3 Marks)

Question 6

- (a) State the importance of voltage control? (3 Marks)
- (b) State briefly how shunt conductance and resistance of conductors cause losses in a power system. (5 Marks)
- (c) List the ways in which fixed losses can be influenced in a power system and list six (6) causes of Non-technical losses (6 Marks)
- (d) What are the factors to be considered for a pole length, pole strength, pole depth and stating the characteristics of a conductor material? (6 Marks)

Question 7

- (a) List the classification of cables (indicating voltage levels) and also list the properties the insulating materials used in cables should have? (4 Marks)
- (b) A 33 kV, 50 Hz, 3-phase underground cable, 4km long uses three single core cables. Each of the conductor has a diameter of 2.5 cm and the radial thickness of insulation is 0.5cm. Determine (i) capacitance of the cables/phase (ii) charging current / phase (iii) total charging kVAR. The relative permittivity of insulation is 3. (6 Marks)
- (c) What are the types of faults that occur in an underground cables and how they can be checked? (6 Marks)
- (d) During the electrical rescue techniques, list the methods of examining a scene and rescuing a victim. (4 Marks)

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATION: Second Semester
 SESSION: 2018/2019
 DATE: October, 2019,



COURSE CODE: EEE 322
 COURSE TITLE: Power system principles
 TIME ALLOWED: 3 Hours

INSTRUCTION: Attempt five (5) questions only.

Question 1

- (a) Briefly explain (i) convenient form (ii) easy control (iii) cleanliness and (iv) high transmission efficiency as electrical energy is superior to all other forms of energy? **(4 Marks)**
- (b) In a generating station which converts heat energy of coal combustion into electrical energy, what is the function of the condenser and list the factors responsible for choice of site for the power plant? **(6 Marks)**
- (c) A hydroelectric station has a mean head of 30m and receives inflows from a catchment area of 300km² having an average rainfall of 200 cm per annum. The rainfall occurs during the monsoon period of 90 days in a year. The storage capacity of the station is limited to half the total inflows. If 70% rainfall reaches the reservoir, determine the minimum capacity of the station. Assume that no head is lost in pipes and the efficiencies of the turbine and generator are 85% and 95% respectively. Determine also the total energy produced and the plant use factor? **(6 Marks)**
- (d) List the factors responsible for comparing the various sources of power generation and the factors responsible for the selection of site for a nuclear power station? **(4 Marks)**

Question 2

- (a) List three (3) advantages and disadvantages of a DC transmission system. **(3 Marks)**
- (b) Fig. 1 is an unconventional energy source for generating electricity, discuss. **(3 Marks)**

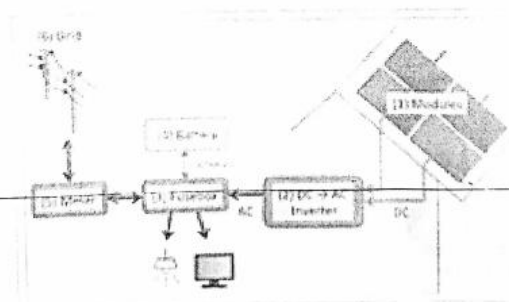


Fig. 1

- (c) What are the types of loads and define the following (i) coincidence factor (ii) load diversity (iii) average load (iv) loss factor **(8 Marks)**
- (d) A generating station has maximum demand of 10,000kW and daily load on the station is as follows:

TIME	KW
11PM – 6AM	2000
6AM – 8AM	3500
8AM – 12NOON	8000
12NOON – 1PM	3000
1PM – 5PM	7500
5PM – 7PM	8500
7PM – 9PM	10000

9PM – 11PM

4500

Choose the size and number of generating units. What reserve plant would be necessary? Find the load factor and plant capacity factor? (6 Marks)

Question 3

- (a) The daily load curves have attained a great importance in generation of electricity supply. list the merits? (7 Marks)
- (b) A generating station is to supply four regions of load whose peak loads are 10 MW, 5 MW, 8 MW and 7 MW. The diversity factor at the station is 1.5 and the average annual load factor is 60%. Calculate:
 (i) the maximum demand on the station,
 (ii) annual energy supplied by the station,
 (iii) Suggest the installed capacity and the number of units. (5 Marks)
- (c) The various problems facing power engineers are considerably reduced by interconnecting different power stations in parallel. list the merits? (3 Marks)
- (d) Fig. 2 presents a typical electrical supply from generator to consumers showing the transformer applications and typical operating voltage levels, briefly give an explanation? (5 Marks)

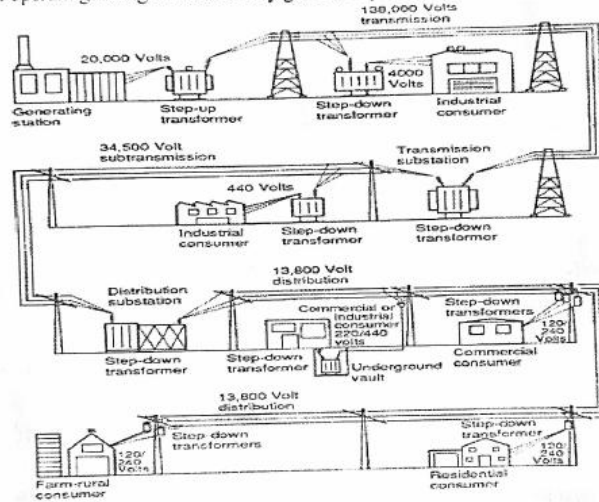


Fig. 2

(5 Marks)

Question 4

- (a) Electricity flows from power plants, through transformers and transmission lines, to substations, distribution lines, and then finally to the electricity consumer. List the principal components of the transmission line as well as the classifications? (6 Marks)
- (b) The distribution system is the electrical system between the sub-station fed by the transmission system and the consumer's meters. With the aid of a diagram discuss the distribution system. (7 Marks)
- (c) Fig. 3 is a type / class of distribution system (d.s), what is this type of d.s and discuss its mode of operation? (3 Marks)



Fig. 3

- (d) List the types of connection systems used for an A.C distribution system as well as methods for obtaining a 3 wire DC system? (4 Marks)

Question 5

- (a) By performance of a transmission line, it means the determination of **regulation** and **efficiency** of the line, define them also state the line parameters governing them? (4 Marks)
- (b) Explain in details the transmission line that has a length 0 – 80km and voltage rating of $\leq 20\text{kV}$ (include phasor diagram). The same transmission line has parameters $R = 0.4\Omega$ and $X = 0.4\Omega$ is delivering 2.000kVA to a load at a pf of 0.8 lagging at the receiving end of the line. If the load voltage is 3.000V, determine the voltage regulation and efficiency of the line. (8 Marks)
- (c) List the factors affecting skin effect in transmission lines and methods for reducing corona discharge effect? (5 Marks)
- (d) List the advantages of using shunt capacitors in a transmission line. (3 Marks)

Question 6

- (a) Define a single line diagram? Fig 4 is a single line diagram of a sample power system, briefly explain the diagram?

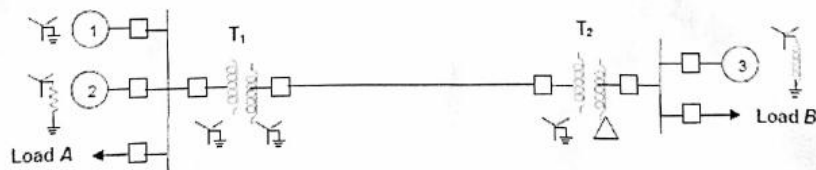


Fig. 4

(4 Marks)

- (b) What is your understanding of a Base kVA? If a transformer is rated 10,000 kVA and has a percentage reactance of 7.5%, what is its percentage reactance when the base kVA is 25000? (4 Marks)
- (c) A single phase transformer in a certain area is rated as 2.5kVA, 11/0.4kV. If the leakage reactance is 0.96Ω when referred to low voltage side, then determine its leakage reactance in per unit. If the same transformer has a base voltage of 1,100 volts, base kVA of 10^6 , what is the base impedance? (8 Marks)
- (d) What are the advantages of per unit (PU) system? (4 Marks)

Question 7

- (a) What is your understanding of transmission and distribution loss, core losses and resistive losses? (6 Marks)
- (b) Explain in details the system parameters shunt conductance and resistance of conductors contribute in losses in transmission lines (where necessary state mathematical formula). (6 Marks)
- (c) What are technical losses and list the types? (3 Marks)
- (d) Explain how step up transformers, transmission line conductors and secondary distribution lines contribute to losses in power system? (5 Marks)

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATION: Second Semester

SESSION: 2016/2017

DATE: August, 2017.



COURSE CODE: EEE 322

COURSE TITLE: Power system principles

TIME ALLOWED: 3 Hours

INSTRUCTION: Attempt five (5) questions only.

Question 1

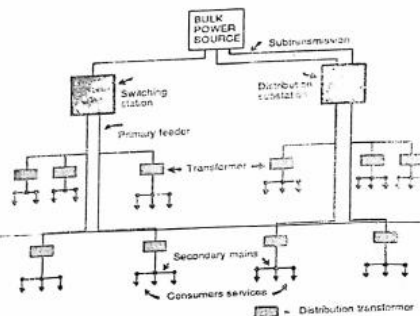
- (a) Electrical energy is superior to all other forms of energy. list the reasons? **(1.5 Marks)**
 (b) State the factors responsible for the selection of site for gas turbine power plants. A diesel power station has the following data:

Fuel consumption/day = 1600 kg
 Units generated/day = 4000 kWh
 Calorific value of fuel = 10,000 kcal/kg
 Alternator efficiency = 96%
 Engine mech. efficiency = 95%

Estimate (i) specific fuel consumption, (ii) overall efficiency, and (iii) thermal efficiency of engine **(6.5 Marks)**

- (c) An hydroelectric station has to operate with a mean head of 50m. It makes use of water collected over a catchment area of 200km² over which the average annual rainfall is 420mm with a 30% loss due to evaporation. Assuming the turbine efficiency as 85% and the alternator efficiency as 80%. Calculate the average power that can be generated. **(6 Marks)**

- (d) State the benefits of interconnection of grids and give an explanation for the distribution system in Fig. 1.

**Fig. 1****(6 Marks)****Question 2**

Define the following terms (i) coincidence factor (ii) load factor (iii) loss factor (iv) coincident demand. List the effects of variable loads on a generating station. **(8 Marks)**

- (b) The following data were collected from the daily load curves of a power station during a year.

Load kW	Duration hours
15,000	87
12,000	And over 876
10,000	And over 1,752

8,000	And over 2,628
6,000	And over 4,380
4,000	And over 7,000
2,000	And over 8,760

Construct the annual load duration curve and find the load factor of the system.

(6 Marks)

(c) What are the advantages of interconnecting generating stations? A base load station having a capacity of 18 MW and a standby station having a capacity of 20 MW share a common load. Find the annual load factors and plant capacity factors of two power stations from the following data:

Annual standby station output = 7.35×10^6 kWh

Annual base load station output = 101.35×10^6 kWh

Peak load on standby station = 12 MW

Hours of use by standby station/year = 2190 hours

(4.5 Marks)

(d) What are the principal elements of a high voltage transmission lines?

(1.5 Marks)

Question 3

(a) What are the classification of overhead transmission lines stating their lengths and voltage levels?

(3 Marks)

(b) Fig.2 and Fig. 3 are types of connection schemes in the distribution system, briefly describe them, stating two (2) advantages each.

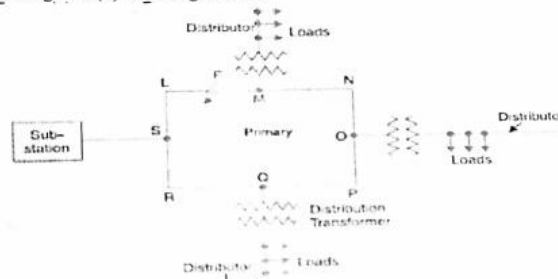


Fig. 2

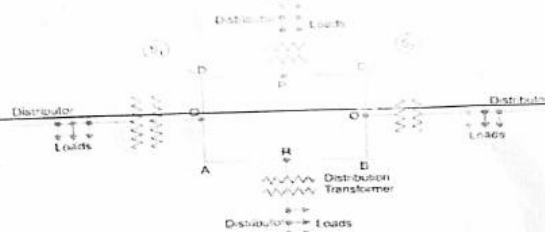


Fig. 3

(8 Marks)

(c) What are the types of connections found in an alternating current (A.C) distribution system and list the requirements of a satisfactory electric supply.

(3 Marks)

(d) An 11 kV, 3-phase transmission line has a resistance of 1.5Ω and reactance of 4Ω per phase. Calculate the percentage regulation and efficiency of the line when a total load of 5000 kVA at 0.8 lagging power factor is supplied at 11 kV at the distant end.

(6 Marks)

Question 4

(a) Define the terms regulation and efficiency of a transmission line. State the parameters governing the performance of transmission lines.

(5 Marks)

- (b) Determine the generalized constants for a transmission for medium lines using nominal T method. (7 Marks)
- (c) Define the term skin effect, stating the factors affecting skin effects and state six (6) methods to reducing corona discharge effects on transmission lines. (4 Marks)
- (d) Estimate the distance over which a load of 15000 kW at a p.f. 0.8 lagging can be delivered by a 3-phase transmission line having conductors each of resistance 1Ω per kilometer. The voltage at the receiving end is to be 132 kV and the loss in the transmission is to be 5%. (4 Marks)

Question 5

- (a) Briefly discuss the single line diagram, impedance diagram and the reactance diagram representation of power system. (6 Marks)
- (b) The one line diagram of a 3-phase power system is shown in Fig. 4. Select a common base of 100MVA and 22 kV on the generator side.

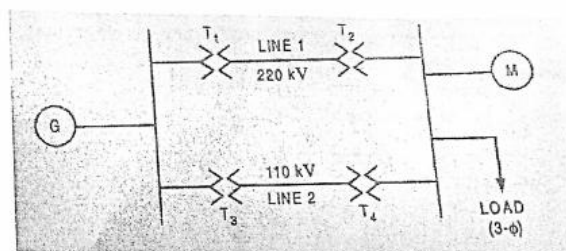


Fig. 4

(8 Marks)

- (c) A single phase 9.6kVA, 500V/1.5kV transformer has an impedance of 1.302Ω with respect to primary side. Find its per unit impedance with respect to primary and secondary sides. (3 Marks)
- (d) Define the term voltage regulation stating the importance. (3 Marks)

Question 6

- (a) With a circuit diagram describe the working operation of a Single Phase Induction Voltage Regulator (4 Marks)
- (b) State briefly how shunt conductance and resistance of conductors cause losses in a power system. (4 Marks)
- (c) List the ways in which fixed losses can be influenced in a power system and list six (6) causes of Non-technical losses (6 Marks)
- (d) What are the following components for the construction of overhead lines used for? (i) vee guards (ii) guard wires (iii) barbed wire. (6 Marks)

Question 7

- (a) List the classification of cables (indicating voltage levels) and also list the properties the insulating materials used in cables should have? (4 Marks)
- (b) A 33 kV, 50 Hz, 3-phase underground cable, 4km long uses three single core cables. Each of the conductor has a diameter of 2.5 cm and the radial thickness of insulation is 0.5cm. Determine (i) capacitance of the cables/phase (ii) charging current / phase (iii) total charging kVAR. The relative permittivity of insulation is 3. (6 Marks)
- (c) What are the types of faults that occur in an underground cables and how they can be checked? (6 Marks)
- (d) During the electrical rescue techniques, list the methods of examining a scene and rescuing a victim. (4 Marks)